

Coordination Before Collision: A Governance Framework for Multi-Agent Clinical AI

Bruce Changlong Xu¹, Ayush Chopra^{2,3}, Alexander J. Ryu⁴

¹Stanford University, Palo Alto, California, USA

²Massachusetts Institute of Technology, Cambridge, Massachusetts, USA

³Project Iceberg at MIT, Cambridge, Massachusetts, USA

⁴Mayo Clinic, Rochester, Minnesota, USA

Correspondence: Alexander J. Ryu

Abstract

Clinical AI systems are beginning to move beyond advice toward multi-step action in scheduling, referral management, documentation, and care coordination. When such systems operate concurrently, harms may arise not only from an individual agent’s output but from interactions across shared patients, workflows, data, and finite resources. Existing healthcare standards and general-purpose agent protocols provide important foundations for data exchange, authorization, tool access, and message transport, yet they do not consistently specify how autonomous clinical actors should expose intent, honor cross-workflow dependencies, coordinate contested resources, or enter a safe mode when coordination fails. We introduce CIRC, Clinical Infrastructure for Reliable Coordination, a conceptual governance framework intended to support requirements analysis and stimulate protocol design, not a proposed standard or validated architecture. CIRC organizes a non-exhaustive taxonomy of agent-local, interaction-and-workflow, and system-and-institutional risks, and maps them to independent governance levers for identity and attribution, scope and authorization, structured intent and shared state, dependency coordination, resource awareness, operational human oversight, and monitoring and failure containment. Deployments may require different combinations of these levers; they do not form a cumulative maturity sequence. We use CliniCARE-Bench as a worked example of how trace-observable agent-local risks can be evaluated without inspecting private model reasoning, while identifying multi-agent interaction and institutional risks that current benchmarks do not test. CIRC therefore offers a starting vocabulary for connecting clinical-agent risks, controls, and evidence requirements before deployment.

1 Introduction

Clinical artificial intelligence (AI) is shifting from decision support toward systems that can select and execute multiple steps in clinical or administrative workflows.[1–3] In this Perspective, a *clinical agent* is a software system that can pursue a delegated objective by choosing and performing workflow actions with less immediate human initiation than conventional decision support. This definition includes administrative agents, supervised clinical agents, and agents permitted to initiate clinically consequential changes; these categories carry different risks and should not be governed identically.

Healthcare’s dominant standards, including FHIR, SMART on FHIR, and CDS Hooks, were designed for data exchange, application authorization, and workflow-triggered decision support.[4–6] These capabilities remain essential and can support portions of agentic workflows through profiles, implementation guides, workflow engines, and local extensions. The governance question is narrower: which additional conventions may be needed when software actors operate continuously, initiate actions, and interact across organizational and vendor boundaries?

Potential failures include stale shared state, actions executed before prerequisites are satisfied, incompatible changes by different agents, retry loops, resource congestion, and escalations with no accountable recipient. We call harms arising from such dependencies *interaction risks*. The term does not imply that every multi-agent failure requires a new protocol: centralized orchestration, EHR-native workflow controls, policy engines, and human-led processes may implement some or all of the needed safeguards.

Single-agent evaluation remains necessary but may be insufficient for systems whose actions interact. Evaluation must distinguish errors in an agent’s own investigation from failures created by shared state, dependencies, handoffs,

resource competition, or institutional response. This distinction motivates a taxonomy that links each risk to observable evidence and candidate controls rather than placing deployments on a single ladder.

We introduce CIRC, Clinical Infrastructure for Reliable Coordination, as a conceptual governance framework intended to stimulate protocol design rather than a proposed standard, implemented architecture, or certification scheme. CIRC makes three contributions. First, it offers a non-exhaustive starting taxonomy spanning agent-local, interaction-and-workflow, and system-and-institutional risks. Second, it defines independent governance levers that institutions can select according to a deployment’s autonomy, clinical consequences, and organizational setting. Third, it connects those risks and controls to evaluation evidence, using an existing trace-based clinical benchmark as a concrete example while making explicit which multi-agent risks remain untested. The levers are neither claimed to be minimal nor sufficient, and future extensions are expected.

2 Why existing infrastructure may be insufficient for clinical agents

Healthcare has invested decades in interoperability infrastructure, including HL7v2 for real-time clinical messaging, FHIR for structured clinical data exchange, SMART on FHIR for application authorization, and CDS Hooks for decision support at defined workflow trigger points.[4–7] These standards are important achievements and could be adapted to support portions of clinical agent coordination. Missing fields, triggers, or scopes may in some cases be addressed through profiles, implementation guides, workflow engines, or vendor extensions. Other gaps concern assumptions about accountable actors, persistent identity, and the context needed to govern autonomous action across systems.

The broader AI ecosystem has begun addressing agent interoperability through protocols such as Google’s Agent-to-Agent protocol, Anthropic’s Model Context Protocol, and IBM’s Agent Communication Protocol.[8–10] A2A and ACP support communication, capability discovery, task management, and message exchange, while MCP standardizes how AI systems connect to tools and data sources. Clinical deployments add constraints that these general protocols do not themselves standardize. Agents may pursue partially conflicting objectives, and some clinical failures cannot be safely handled through retry because downstream actions may be difficult to reverse. The central problem is therefore not only capability discovery but capability constraint: what an agent may do, for which patients, under which conditions, with which approvals, and how it must coordinate when its actions intersect with others.

Existing AI risk frameworks characterize what can go wrong. The AI Risk Repository synthesizes more than 700 risks across 43 taxonomies, and the NIST AI Risk Management Framework organizes governance functions for trustworthy systems.[11, 12] Clinical agent benchmarks evaluate whether individual agents perform correctly in realistic environments.[13, 14] Multi-agent clinical deployment additionally raises questions about interacting agents acting concurrently on shared patients, finite resources, and institutional workflows. Four gaps illustrate this distinction.

Insufficient agent observability and attribution

As agents move from recommending actions to initiating orders, updating documentation, routing referrals, or modifying schedules, audit infrastructure must distinguish human-authored actions, human-approved agent actions, and autonomous agent-initiated actions. Current mechanisms are commonly organized around user accounts, applications, service credentials, and record modifications rather than persistent agent identity across vendors and institutions. FHIR Provenance can represent that a system, practitioner, organization, or device participated in an activity, but implementations do not generally require a cross-vendor mechanism for cryptographically verifiable agent identity, model version, authorization context, and cross-organizational persistence. Without this context, adverse-event review may struggle to reconstruct which agent acted, under whose authority, and within which scope.

Missing intent and dependency protocols

Clinical coordination requires expressing dependencies across disease-focused, patient-focused, capability-focused, and operational agents. A diabetes-management agent, oncology agent, record-summarization agent, and imaging-scheduling agent may each impose constraints on one care pathway. These constraints can include prerequisites, timing windows, resource needs, escalation rules, data-access limits, and reversal obligations. Healthcare standards can exchange data and invoke workflow hooks, while general-purpose agent protocols can communicate and hand

off tasks. Neither category, by itself, standardizes a clinical dependency protocol that obliges agents to declare how a proposed action affects downstream workflows and patient state before execution.

Capability boundaries scoped to human roles

Healthcare defines role-based authority around expertise, licensure, institutional policy, and accountability. Agents require analogous constraints, potentially with broad reading privileges but strict acting privileges. SMART on FHIR provides application-level authorization to data and services, but these scopes do not encode the complete clinical action context of autonomous behavior. A diabetes agent might be allowed to recommend insulin adjustments in one setting, require pharmacist approval in another, and be prohibited from acting autonomously in a third. Clinical deployments therefore may need enforceable declarations of what an agent can do, for whom, under which conditions, and through which escalation path.

Limited system-level resource coordination

When many agents independently conclude that the same time window is optimal for imaging, infusion, discharge planning, or specialist review, they could collectively overwhelm shared resources even if each decision is locally reasonable. Prior work calls this type of convergence a “digital stampede.”[15] Historical utilization statistics and single-agent benchmarks may not expose risks arising dynamically from the interaction of agents, patients, resources, and workflows. Existing standards can represent many relevant facts, and workflow engines can coordinate specified processes, but aggregate demand signals and population-level stress tests are not routinely standardized for clinical agent ecosystems.

3 A non-exhaustive taxonomy of clinical-agent risk

CIRC begins with risks rather than levels. The taxonomy is intentionally non-exhaustive: it is a starting point for requirements analysis, not a claim to enumerate every hazard or rank deployments on a maturity scale. We organize risks by the locus at which evidence and control are needed (Table 1). A single event can occupy more than one domain, and risk domains do not imply causal independence.

Table 1: A non-exhaustive taxonomy of clinical-agent risks.

Risk domain	Illustrative risks	Primary unit of evaluation
Agent-local evidence and reasoning	Wrong-patient or wrong-episode attribution; temporal misalignment; substitution of codes or orders for clinical evidence; incomplete evidence retrieval; rule misapplication; unsupported inference; false closure under missing or conflicting evidence.	One agent’s observable investigation, actions, report, and abstention behavior.
Interaction and workflow	Stale or inconsistent shared state; unmet prerequisites; incompatible or duplicated actions; failed handoffs; race conditions; incorrect dependency graphs; recursive retry or negotiation loops.	Concurrent trajectories around a shared patient, task, or care pathway.
System and institutional	Resource congestion; correlated failure across patients; unavailable or misleading aggregate signals; escalation without an accountable responder; coordination-layer outage; cross-organizational trust failure; alert burden, inequity, and drift.	Agent populations, shared resources, institutions, and human response processes over time.

The first domain is already visible in emerging clinical-agent evaluations. ClinCARE-Bench, for example, evaluates retrospective clinical audits over 25 scenarios and 750 patient-specific cases in a governed, replayable EHR environment.[16] It separately scores final verdicts, patient-evidence grounding, policy use, process adherence, and calibrated abstention. Across 16 evaluated systems, reported four-way accuracy ranged from 65.3% to 76.1%, while accuracy that also excluded prohibited process shortcuts was 4.8 to 14.8 percentage points lower. Every

evaluated system over-committed more often than it over-abstained. These findings do not validate CIRC and do not test multi-agent coordination. They demonstrate something narrower and useful: governance-relevant properties can be specified against observable traces, and a correct output can conceal a defective investigation.

Interaction and system risks require different evidence. A benchmark in which one agent reviews one record cannot measure cross-agent deadlock, conflicting modifications, population-level demand, or whether a clinician responds to an escalation. These risks require multi-agent replay, workflow stress tests, sociotechnical simulation, prospective shadow deployment, or post-deployment monitoring. CIRC uses the taxonomy to make those evidence boundaries explicit rather than treating single-agent accuracy as a proxy for system safety.

4 CIRC governance levers

CIRC maps risks to independent governance levers (Table 2). A lever is a capability or institutional control that can be implemented through a protocol, a centralized service, an EHR-native workflow, or another architecture. Deployments may implement the levers in different orders and at different strengths. None is asserted to be individually necessary, collectively sufficient, or unique to CIRC.

Table 2: Independent CIRC governance levers and the questions they make answerable.

Governance lever	Core question	Illustrative evidence
Identity and attribution	Which agent and version acted, under whose authority, and in which execution context?	Signed identity, delegation chain, provenance record, immutable audit event.
Scope and authorization	Was this action permitted for this patient, task, setting, and degree of autonomy?	Machine-readable capability and scope declaration; policy decision and override record.
Structured intent and shared state	What action is proposed, against which state version, with what time limit and reversal plan?	Typed intent message, state version, expected effect, expiry, and idempotency key.
Dependency coordination	Which prerequisites and downstream obligations must hold before and after the action?	Dependency graph, prerequisite checks, acknowledgement, cancellation propagation.
Resource and population awareness	Could locally reasonable actions create congestion or correlated failure?	Privacy-preserving capacity, queue, demand, priority, and concentration signals.
Operational human oversight	Who must respond, by when, with what authority, and what happens after non-response?	Named role, response deadline, fallback recipient, override, suspension, and closure.
Monitoring and failure containment	How are faults detected, bounded, revoked, and handled during degraded operation?	Health state, replay protection, rate limits, circuit breakers, safe mode, and recovery log.

The mapping is many-to-many. Dependency coordination can mitigate both agent-local omissions and cross-agent prerequisite violations. Identity supports auditability but does not, by itself, prevent an authorized agent from acting on stale state. Human oversight can contain uncertainty or conflict only when escalation has an accountable recipient, response window, and fallback behavior. Resource signals may reduce congestion while introducing privacy, manipulation, and allocation risks of their own.

This framing is compatible with multiple architectures. A central workflow engine may maintain the dependency graph; an institutional policy engine may enforce scope; an EHR-native service may mediate orders; a supervisory agent may detect conflict; and direct agent-to-agent messages may carry intent across vendors. Protocol-level fields are most useful where information must cross those boundaries, but CIRC does not assume that every control belongs in a new protocol or that decentralized agent communication is preferable to orchestration.

5 From risk taxonomy to evaluation

The taxonomy separates three questions that were conflated in the submitted simulations. First, can a risk be observed? Candidate measures include attribution completeness, stale-state reads, unresolved dependencies, duplicated or conflicting actions, retry amplification, queue concentration, and unowned escalations. Second,

Risk locus	Governance task	Candidate levers
Agent-local	Establish whether evidence, timing, policy, scope, and uncertainty support an action.	Identity; scope; structured state; oversight
Interaction and workflow	Preserve shared state and dependencies as actions cross agents and systems.	Structured intent; dependency coordination; containment
System and institutional	Detect correlated demand and maintain accountability during escalation or outage.	Resource awareness; oversight; monitoring and containment

Figure 1: CIRC connects risk loci to governance tasks and candidate controls. Arrows are intentionally not one-to-one: controls overlap, and the appropriate implementation depends on the deployment.

can a control detect or contain the risk without blocking valid work? Matched stress tests should report both fault containment and preservation of unaffected workflows. Third, does the control improve clinical or operational outcomes? That question requires prospective evaluation and cannot be inferred from a deterministic demonstration.

Trace-based benchmarks such as CliniCARE provide a concrete template for the first question. They log retrieval, computation, policy access, citations, and reports, allowing process defects to be separated from final-answer errors without inspecting private reasoning. Extending this approach to interaction risks would require linked traces from multiple agents acting on shared state, together with prespecified fault models, applicable denominators, baselines, repeated runs, uncertainty estimates, and failure cases. Population-level resource tests would additionally need realistic arrival processes, capacity constraints, and human response models.

Accordingly, this Perspective does not present the removed toy simulations as evidence of control effectiveness. Instead, it proposes a staged evaluation agenda: trace-based assessment of agent-local risks; matched multi-agent replay for interaction and workflow risks; system stress testing for shared resources and common-mode failure; prospective shadow deployment; and monitored clinical use where regulatory and institutional requirements permit.

6 Limitations and future directions

CIRC is a conceptual and non-exhaustive framework. It does not specify a complete technical implementation, establish the necessity or sufficiency of any lever, certify deployment readiness, or demonstrate clinical benefit. The taxonomy will require extension as clinical agents acquire new capabilities and as evidence emerges from real workflows. Its categories also overlap: a stale-state error, for example, can reflect an agent-local retrieval failure, an interaction failure, and an institutional monitoring failure.

CliniCARE illustrates trace-observable agent-local risks, but it is a retrospective single-agent audit benchmark rather than a test of multi-agent coordination. Its findings cannot support claims about cross-vendor messaging, shared-resource control, human response, or prevented patient harm. Likewise, population-level clinical prediction is outside the scope of a coordination framework and must be developed and validated as a separate clinical model.

Implementation questions remain about identity issuance and revocation, consistency guarantees, message loss and replay, network partitions, privacy-preserving aggregation, and safe behavior during coordination-layer outage. Governance questions include liability, regulatory classification, institutional autonomy, local configuration, alert burden, and equitable access. These issues will play out differently under regimes such as the FDA’s device-centered framework and the European Union’s risk-tiered AI Act.[17, 18] Future work should compare protocol, orchestration, policy-engine, EHR-native, and human-led implementations against common risk-specific endpoints.

7 Conclusion

Multi-agent clinical safety cannot be inferred from individual-agent performance alone. The relevant risks span local investigations, interactions across shared workflows, and the institutions that authorize, monitor, and respond to autonomous action. CIRC offers a starting taxonomy for those risks and a set of independent governance levers that can be implemented through different technical architectures.

The framework’s value should be judged by whether it makes requirements and evidence more precise, not by treating its categories as a maturity ladder or its controls as established safety invariants. Existing trace-based benchmarks show how some agent-local risks can be measured. The next challenge is to build equally inspectable evaluations for shared state, dependencies, resource competition, degraded operation, and human response before autonomous clinical agents operate at scale.

Funding

This work was accomplished without funding.

Competing interests

The authors declare no competing interests.

References

- [1] Michael Moritz, Eric Topol, and Pranav Rajpurkar. Coordinated ai agents for advancing healthcare. *Nature Biomedical Engineering*, 9:432–438, 2025. doi: 10.1038/s41551-025-01363-2.
- [2] Andrew A. Borkowski and Alon Ben-Ari. Multiagent ai systems in health care: Envisioning next-generation intelligence. *Federal Practitioner*, 42:188–194, 2025. doi: 10.12788/fp.0589.
- [3] R. Andrew Taylor. Ai agents, automaticity, and value alignment in health care. *NEJM AI*, 2(8), 2025. doi: 10.1056/AIp2401165.
- [4] HL7 International. Fast healthcare interoperability resources (fhir), release r4, 2019. URL <https://www.hl7.org/fhir/>.
- [5] HL7 International. Cds hooks specification, release 1.0, 2018. URL <https://cds-hooks.hl7.org>.
- [6] Joshua C. Mandel, David A. Kreda, Kenneth D. Mandl, Isaac S. Kohane, and Rachel B. Ramoni. Smart on fhir: a standards-based, interoperable apps platform for electronic health records. *Journal of the American Medical Informatics Association*, 23:899–908, 2016. doi: 10.1093/jamia/ocv189.
- [7] Health Level Seven International. Hl7 version 2.5.1: Application protocol for electronic data exchange in healthcare environments, 2007. URL https://www.hl7.org/implement/standards/product_brief.cfm?product_id=144.
- [8] Rao Surapaneni, Mukul Jha, Matt Vakoc, and Todd Segal. Announcing the agent2agent protocol, 2025. URL <https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/>.
- [9] Anthropic. Introducing the model context protocol, 2024. URL <https://www.anthropic.com/news/model-context-protocol>.
- [10] Samuel Besen and Anna Gutowska. What is agent communication protocol?, 2025. URL <https://www.ibm.com/think/topics/agent-communication-protocol>.
- [11] Peter Slattery et al. The ai risk repository: A meta-review, database, and taxonomy of risks from artificial intelligence. *arXiv preprint arXiv:2408.12622*, 2024. doi: 10.48550/arXiv.2408.12622.
- [12] National Institute of Standards and Technology. Artificial intelligence risk management framework (ai rmf 1.0). Technical Report NIST AI 100-1, NIST, 2023.
- [13] Yixing Jiang et al. Medagentbench: A virtual ehr environment to benchmark medical llm agents. *NEJM AI*, 2(9), 2025. doi: 10.1056/AIdbp2500144.
- [14] Dyke Ferber et al. Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology. *Nature Cancer*, 6:1337–1349, 2025. doi: 10.1038/s43018-025-00991-6.

- [15] Ayush Chopra et al. Ripple effect protocol: Coordinating agent populations. *arXiv preprint arXiv:2510.16572*, 2025. doi: 10.48550/arXiv.2510.16572.
- [16] Veronica Chatrath, Bryan Zhu, George Pu, et al. Clinicare-bench: Clinical calibrated audit of medical reasoning in ehr. Scale AI Research, 2026. URL <https://labs.scale.com/papers/clinicare-bench>.
- [17] U.S. Food and Drug Administration. Artificial intelligence/machine learning-based software as a medical device action plan. Technical report, FDA, 2021. URL <https://www.fda.gov/media/145022/download>.
- [18] European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>.
- [19] J. Y. Kim, S. J. Nam, J. E. Lee, et al. Real-world evidence of neoadjuvant docetaxel/carboplatin/trastuzumab/pertuzumab in patients with her2-positive early or locally advanced breast cancer. *Cancer Research and Treatment*, 54:1091–1098, 2022. doi: 10.4143/crt.2021.901.
- [20] C. Hockaday, J. Chiramel, et al. Efficacy and safety of neoadjuvant docetaxel, carboplatin, trastuzumab and pertuzumab in patients with her2-positive early breast cancer. *Clinical Oncology*, 2024. doi: 10.1016/j.clon.2024.103717.
- [21] J. A. W. Polderman, J. Hermanides, and A. H. Hulst. Update on the perioperative management of diabetes mellitus. *BJA Education*, 24:261–269, 2024. doi: 10.1016/j.bjae.2024.04.007.
- [22] Centre for Perioperative Care. Guideline for the perioperative care of people with diabetes mellitus undergoing elective and emergency surgery, 2022. URL <https://cpoc.org.uk/guidelines-resources/guidelines>.
- [23] Alexander R. Lyon et al. 2022 esc guidelines on cardio-oncology. *European Heart Journal – Cardiovascular Imaging*, 23:e333–e465, 2022. doi: 10.1093/ehjci/jeac106.
- [24] Alana F. Yu et al. Cardiac safety of reduced cardiotoxicity surveillance during her2-targeted therapy. *JACC: CardioOncology*, 7:430–441, 2025. doi: 10.1016/j.jacc.2025.05.006.
- [25] David J. Kuter. Treatment of chemotherapy-induced thrombocytopenia in patients with non-hematologic malignancies. *Haematologica*, 107:1243–1263, 2022. doi: 10.3324/haematol.2021.279512.